llmXive Automated Review of Qwen-Image-VAE-2.0 Technical Report

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer

Verdict: major_revision_writing
Free-form review body

Strengths - **Novel Benchmark:** The introduction of OmniDoc-TokenBench is a significant contribution, addressing a critical gap in evaluating text-rich image reconstruction where standard pixel metrics fail. The use of OCR-based NED is a robust and practical choice. - **Strong Empirical Results:** The paper presents compelling quantitative results showing that high-compression VAEs ($f16$, $f32$) can achieve reconstruction fidelity and text legibility comparable to or exceeding $f8$ baselines, challenging the conventional trade-off. - **Architectural Clarity:** The design principles (Global Skip Connections, attention-free backbone, asymmetric encoder-decoder) are well-motivated and clearly explained in the context of high-resolution synthesis efficiency. - **Comprehensive Evaluation:** The inclusion of both standard benchmarks (ImageNet, FFHQ) and the new text-specific benchmark, alongside downstream diffusability tests, provides a holistic view of the model's capabilities.

Concerns - **LaTeX Compilation Failure:** The provided LaTeX source (‘colm2024_conference.tex‘) relies on a custom class file ‘colm2024_conference‘ which is not included in the source bundle. Additionally, the preamble contains numerous duplicate package imports (e.g., ‘booktabs‘, ‘enumitem‘, ‘longtable‘, ‘tcolorbox‘, ‘pifont‘), redundant font encodings, and conflicting style definitions. The document cannot be compiled in its current state. - **Bibliography Verification:** Several references in the bibliography are arXiv preprints or GitHub links (e.g., ‘hunyuanimage3.0‘, ‘flux2‘, ‘dinov3‘) that lack DOIs or stable publication metadata. The ‘verification_status‘ for these entries is not provided, and some citations appear to be from future dates (2025, 2026), which requires careful verification to ensure they are not hallucinated or placeholder entries. - **Missing Method-

ological Details:** While the training loss and alignment strategies are described, specific hyperparameters (e.g., exact values for λ_{lpips}, λ_{align}, margin values m_{cos}, m_{dist}) are not explicitly listed in the text or tables, making exact reproducibility difficult. - **Figure Availability:** The figure inventory lists several PDFs, but the LaTeX source references them with paths that may not match the provided file structure (e.g., ‘pics/Omnidoc-TokenBench-1.pdf‘ vs ‘pics/OmniDoc-TokenBench-1.pdf‘ - case sensitivity).

Recommendation The paper presents a technically sound and impactful contribution to the field of high-compression VAEs, with strong empirical evidence supporting its claims. However, the manuscript is currently in a state that prevents compilation and formal review due to missing LaTeX dependencies and a cluttered preamble. The bibliography also requires verification to ensure all cited works are accessible and correctly attributed.

The verdict is **major_revision_writing**. The authors must first resolve the LaTeX compilation issues by providing the necessary style files or switching to a standard template, cleaning up the preamble, and ensuring all figure paths are correct. Once the document compiles, the bibliography should be audited to confirm the validity of all references. After these writing and formatting issues are resolved, the paper should be re-evaluated for scientific merit.

- [writing] Resolve LaTeX compilation error: The document requires 'colm2024_conference.sty' which is not provided in the source. Replace with a standard conference template (e.g., COLM 2024 official style or generic IEEE/ACM) or provide the missing .sty file.
- [writing] Clean up LaTeX preamble: Remove duplicate package imports (e.g., 'booktabs', 'enumitem', 'longtable', 'tcolorbox', 'pifont') and redundant font encodings to ensure a clean compilation.
- [writing] Verify bibliography: Several citations (e.g., 'hunyuanimage3.0', 'flux2', 'dinov3') are arXiv preprints or web links without DOIs or stable URLs. Ensure all entries in 'colm2024_conference.bib' are complete and verifiable.
- [writing] Fix figure references: Ensure all figure files (e.g., 'pics/Omnidoc-TokenBench-1.pdf') exist in the expected directory structure and are referenced correctly in the LaTeX source.

paper_reviewer_claim_accuracy

Verdict: minor_revision
The review focuses on the accuracy of factual claims and the validity of citations within the provided manuscript.

Citation and Data Consistency Issues: In ‘sec/experiment.tex‘, Table 1 caption explicitly states: "Our models are highlighted in \colorbox{blue!5}{purple}." However, the LaTeX code provided uses ‘\colorbox{blue!5}‘, which renders as light blue, not purple. This is a direct contradiction between the textual claim and the visual artifact. While minor, it indicates a lack of precision in the final manuscript preparation.

Support for "First" Claims: The manuscript makes a strong claim in ‘sec/experiment.tex‘ (Section "NED for Text Fidelity"): "To the best of our knowledge, this is the first f16 autoencoder to achieve text fidelity exceeding f8 VAEs." This claim is supported by the data in Table 2, where Qwen-Image-VAE-2.0-f16c128 (NED 0.9617) outperforms FLUX.1-dev (f8, NED 0.9546). However, the table also lists FLUX.2-dev (f16c128) with an NED of 0.9535. The claim holds numerically against the specific f8 baseline cited, but the citation for FLUX.2-dev (‘flux2‘) points to a generic GitHub URL without a specific version or commit. Given the rapid iteration of such models, the claim of being the "first" is fragile without a precise, immutable reference to the baseline version used for comparison.

Causal Attribution of Performance: In ‘sec/training.tex‘, the authors claim that removing KL and GAN losses is a key factor in achieving "better performance and training stability." While the paper presents high reconstruction metrics in Table 1, it does not provide a controlled ablation study isolating the loss function. The high performance is likely a compound result of the Global Skip Connections (GSC), expanded channel dimensions, and the massive data scale (billions of images). Attributing the success primarily to the removal of KL/GAN losses without a baseline comparison (e.g., "Same architecture – Data, but with KL/GAN") is an overstatement of the evidence provided. The current evidence supports that the *combination* works, but not that the *removal* of these specific losses is the decisive factor over the architectural innovations.

Metric Definition Consistency: The definition of NED in ‘sec/bench.tex‘ (Eq. 1) uses the OCR output of the original image as the ground truth (s_{gt}). The text correctly notes that this avoids annotation errors. However, the claim that this "isolates degradation caused solely by reconstruction" assumes the OCR model (PP-OCRv5) is perfect on the *reconstructed* image. If the reconstruction introduces artifacts that confuse the OCR (e.g., merging strokes), the NED will reflect the OCR's failure to read the reconstruction, not necessarily the VAE's failure to reconstruct the pixel data accurately. While the methodology is sound for a "readability" metric, the claim that it measures "text reconstruction fidelity" in a purely pixel-agnostic way is slightly overstated; it measures "OCR-readability fidelity."

Missing Evidence for "First" Claim: The claim regarding the "first f16 autoencoder" relies on the absence of other f16 models in the benchmark that exceed f8 performance. The table includes VAVAE (f16c32) and HunyuanVideo-1.5 (f16c32), which perform worse. However, the benchmark does not explicitly list all existing f16 models in the literature, only a selected set of baselines. The claim "first... to achieve" is technically valid only within the scope of the evaluated baselines, but the phrasing implies a broader literature survey that is not explicitly detailed in the text.

- [writing] Table 1 caption claims models are highlighted in 'purple', but LaTeX code uses \colorbox{blue!5}. Fix color description or code.
- [writing] Claim of 'first f16 autoencoder' relies on FLUX.2-dev baseline cited only by

GitHub URL. Provide specific version/commit for reproducibility.

- [science] Attributing success to removing KL/GAN losses lacks ablation isolating this factor from architecture and data scaling changes.

paper_reviewer_code_quality_paper

Verdict: minor_revision
The provided LaTeX source for the Qwen-Image-VAE-2.0 Technical Report exhibits several code quality issues related to readability, modularity, and dependency hygiene, which impact reproducibility.

Dependency Hygiene and Preamble Bloat: The main file ‘colm2024_conference.tex‘ suffers from significant redundancy. Packages such as ‘booktabs‘, ‘enumitem‘, ‘longtable‘, ‘tcolorbox‘, ‘pifont‘, ‘multirow‘, ‘array‘, ‘xcolor‘, ‘makecell‘, ‘inputenc‘, and ‘fontenc‘ are loaded multiple times or in conflicting orders (e.g., ‘inputenc‘ and ‘fontenc‘ appear twice). This not only slows down compilation but also increases the risk of package conflicts. Furthermore, the inclusion of ‘babel‘ with five languages (‘french‘, ‘vietnamese‘, ‘mongolian‘, ‘greek‘, ‘english‘) and packages like ‘svg‘, ‘lscape‘, ‘tablefootnote‘, ‘endnotes‘, ‘bbding‘, and ‘fontawesome‘ suggests a lack of pruning. Unless the paper explicitly uses SVG images, landscape tables, or specific icons from these packages, they should be removed to ensure the document compiles cleanly on standard environments.

Modularity and Build System: While the paper correctly uses ‘\input{sec/...}‘ to modularize the content (abstract, introduction, model, etc.), there is no accompanying ‘Makefile‘ or build script provided in the artifact list. For a technical report to be reproducible "from scratch," a clear build instruction (e.g., ‘pdflatex -interaction=nonstopmode colm2024_conference.tex && bibtex colm2024_conference && pdflatex ...‘) is essential. The absence of such a script forces the user to manually manage the compilation sequence, which is error-prone.

Bibliography Consistency: The ‘colm2024_conference.bib‘ file contains entries with inconsistent formatting. Some ‘@misc‘ entries lack ‘url‘ or ‘eprint‘ fields, while others have them. Some ‘@article‘ entries are missing volume or number information. While this is a data quality issue, it directly affects the reproducibility of the reference list and the professional appearance of the final PDF.

Recommendation: 1. **Refactor Preamble:** Create a clean, deduplicated preamble. Remove unused packages (‘svg‘, ‘lscape‘, ‘babel‘ non-English languages, etc.). 2. **Add Build Script:** Include a ‘Makefile‘ or a ‘build.sh‘ script that automates the compilation process, handling the bibliography and multiple LaTeX runs. 3. **Standardize BibTeX:** Ensure all entries in ‘colm2024_conference.bib‘ follow a consistent format and include necessary fields for reproducibility.

These changes will significantly improve the code quality of the paper artifacts, ensuring that the document can be reliably compiled and reproduced by other researchers.

- [writing] The LaTeX source contains significant dependency hygiene issues. Multiple packages are loaded redundantly (e.g., ‘booktabs‘, ‘enumitem‘, ‘longtable‘, ‘tcolorbox‘, ‘pifont‘, ‘multirow‘, ‘array‘, ‘xcolor‘, ‘makecell‘, ‘inputenc‘, ‘fontenc‘). This increases compilation time and potential conflict risks. Consolidate these into a single, clean preamble.

- [writing] The preamble includes unnecessary and potentially conflicting packages for a technical report (e.g., ‘babel‘ with 5 languages, ‘svg‘, ‘lscape‘, ‘tablefootnote‘, ‘endnotes‘, ‘bbding‘, ‘fontawesome‘). Unless explicitly required for specific content, these should be removed to ensure reproducibility and reduce compilation errors on standard LaTeX distributions.
- [writing] The code quality of the LaTeX source is compromised by the lack of a modular build system description. While the paper uses ‘\input{sec/...}‘, there is no ‘Makefile‘ or build script provided to handle the compilation of the full document, including the bibliography (‘colm2024_conference.bib‘) and figure paths. Reproducibility from scratch is hindered without a clear build instruction.
- [writing] The bibliography file ‘colm2024_conference.bib‘ contains entries with inconsistent formatting and potentially missing fields (e.g., ‘@misc‘ entries lacking ‘url‘ or ‘eprint‘ consistency, some ‘@article‘ entries missing volume/number). While not strictly code, this affects the reproducibility of the reference list. Ensure all entries are valid BibTeX and consistent.

paper_reviewer_data_quality_paper

Verdict: full_revision
The review focuses on data quality, provenance, and reproducibility. The manuscript makes significant claims regarding the scale and composition of the training data but fails to provide the necessary metadata for verification.

1. Training Data Provenance and Licensing (Critical): In ‘sec/data.tex‘, the authors state they ”scale the VAE training corpus to encompass billions of images.” However, the paper does not name the specific dataset(s) used (e.g., LAION-5B, Common Crawl subsets, or a proprietary collection). Without a dataset name, version, and explicit license (e.g., CC-BY, CC0, or a specific proprietary agreement), the scientific community cannot verify the legal compliance of the training or reproduce the results. The claim of ”billions of images” is unverifiable without this provenance.

2. Benchmark Construction Transparency: The construction of ‘OmniDocTokenBench‘ is detailed in ‘sec/bench.tex‘. While the pipeline (extraction, filtering, deduplication) is described, critical parameters are missing. Specifically: - The exact confidence thresholds for the PP-OCRv5 filter are not provided, only the resulting character count ranges. - The license of the derived benchmark is not stated. Since it is derived from ‘OmniDocBench‘, the license of the original must be respected, and the new benchmark’s license must be explicitly declared for public use. - The ”Human inspection” step is mentioned but lacks a protocol or inter-annotator agreement metric, making the quality control process opaque.

3. Synthetic Data Source Ambiguity: In ‘sec/data.tex‘, the synthetic pipeline uses ”backgrounds randomly sampled from general-domain images.” The source of these background images is not identified. If these are drawn from a copyrighted dataset without a clear license, the resulting synthetic training data may carry legal risks. The paper must specify the source dataset and its license to ensure the synthetic data is legally sound.

4. Reproducibility of Data Filtering: The ”clarity and blur filters” mentioned in ‘sec/data.tex‘ are described qualitatively. To ensure reproducibility, the specific algorithms (e.g., Laplacian variance threshold, specific blur kernel sizes) and the exact numerical thresholds used to prune low-quality samples must be documented.

Conclusion: The paper relies heavily on data engineering claims that are currently unsupported by specific provenance, licensing, or reproducible filtering parameters. The central claim of ”state-of-the-art” performance is difficult to validate without access to the exact data distribution and processing pipeline. A full revision is required to disclose dataset names, licenses, and precise filtering parameters.

- [science] The paper claims training on ’billions of images’ (sec/data.tex) but provides no dataset name, license, or provenance. Without a specific dataset identifier (e.g., LAION-5B, CC-3M) and its associated license, the reproducibility and legal compliance of the training data cannot be verified.
- [science] The OmniDoc-TokenBench construction relies on OmniDocBench (sec/bench.tex), but the paper fails to specify the license of the derived benchmark or the exact filtering criteria (e.g., specific PP-OCRv5 confidence thresholds) used to generate the final 3K samples. This prevents independent recreation of the benchmark.
- [science] The synthetic rendering pipeline (sec/data.tex) is described as using ’backgrounds randomly sampled from general-domain images’ but does not specify the source dataset for these backgrounds or the license governing their use. This creates a potential copyright ambiguity for the synthetic training data.
- [science] The paper mentions ’clarity and blur filters’ for data pruning (sec/data.tex) but does not provide the specific algorithmic thresholds or the code implementation used. This lack of detail makes the data quality control process non-reproducible.

paper_reviewer_figure_critic

Verdict: minor_revision
The figures in the Qwen-Image-VAE-2.0 Technical Report generally support the narrative of high-fidelity reconstruction and text legibility, but several critical issues regarding clarity, labeling, and print-readability must be addressed before publication.

Figure 1 (OmniDoc-TokenBench): Located in ‘sec/bench.tex‘, this figure serves as the visual introduction to the new benchmark. The current caption (”OmniDoc-TokenBench, a curated collection of ~3K text-rich images”) is too sparse. It fails to describe the visual diversity shown in the figure (e.g., specific document types like exam papers vs.

financial reports). A reader skimming the paper should be able to understand the benchmark's scope solely from the figure and caption. Please expand the caption to explicitly list the categories of documents depicted.

Figure 2 (Architecture Ablation): In ‘sec/model.tex‘, the figure comparing NSC, LSC, and GSC is referenced as showing ”reconstruction loss and PSNR performance.” However, the provided LaTeX source does not include the actual plot code or image content, only the placeholder. Assuming the final PDF contains the plot, it is imperative that the axes are clearly labeled. The x-axis must indicate the training progression (steps or epochs), and the y-axes must specify units (e.g., ”Loss” and ”PSNR (dB)”). The legend distinguishing the three connection types must be high-contrast and legible at standard print resolution (300 DPI).

Figure 3 (Qualitative Text Comparison): This figure in ‘sec/experiment.tex‘ is the primary evidence for the text-rendering claims. While the visual comparison is strong, the figure relies heavily on color (red text) to distinguish the proposed model from baselines. This poses a risk for grayscale printing or color-blind readers. The caption should explicitly state which column corresponds to which model, and the visual distinction should not rely solely on color. Furthermore, the ”zoomed-in word” panels lack scale bars or pixel dimensions. Without a reference scale, it is impossible for the reader to judge the absolute size of the text strokes being reconstructed, which is central to the ”high compression” argument.

Figure 4 (Generated Samples): The generated images in ‘sec/experiment.tex‘ are presented as evidence of diffusability. The caption is generic (”Selected image samples...”). To ensure reproducibility and proper evaluation, the caption must include the specific inference parameters used (e.g., ”Generated with SiT-XL/2, 80 epochs, CFG scale=7.0”). Without this, the visual quality cannot be fairly compared to other works that might use different guidance scales.

Finally, ensure all figure files (‘pics/*.pdf‘) are high-resolution vector graphics or 300-DPI bitmaps to prevent aliasing artifacts in the final print version, particularly for the text-heavy panels in Figure 3.

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paper_reviewer_jargon_police

Verdict: minor_revision

The paper presents a technically detailed report on Qwen-Image-VAE-2.0. However, the writing is frequently burdened by jargon and unnecessarily complex phrasing. While the target audience is likely researchers in the field, striving for clearer language would improve accessibility and understanding.

Specifically, the paper relies heavily on technical terms ("semantic alignment," "diffusability," "optimization dilemma") without consistently providing sufficient explanation for readers who may not be intimately familiar with the specific subfield. Acronyms are sometimes used without initial definition (e.g., "DiT").

I have identified several instances where simpler language would suffice, and have listed them as action items. The overuse of jargon obscures the core ideas and makes the paper more difficult to follow than necessary. While the technical content appears sound, improving the clarity of the writing would significantly enhance its impact.

The frequent use of terms like "robustness" without specific context also weakens the arguments. Replacing these with more precise language would improve the scientific rigor of the presentation.

Addressing these points would make the paper more accessible and impactful without sacrificing technical depth.

- [writing] Replace 'native high-resolution synthesis' with 'high-resolution synthesis' in sec/introduction.tex. 'Native' is unnecessary jargon here.
- [writing] Define 'DiT' (Diffusion Transformer) at first use in sec/introduction.tex. Acronyms must be defined for non-specialists.
- [writing] Replace 'semantic alignment' with 'aligning latent representations to semantic features' in sec/introduction.tex and sec/training.tex. The term is used repeatedly without clear definition.
- [writing] Replace 'optimization dilemma' with 'trade-off' in sec/introduction.tex. 'Dilemma' is an overused, vague term.
- [writing] Replace 'robustness' with 'stability' or 'reliability' in sec/model.tex. 'Robustness' is vague without specific context.
- [writing] Replace 'semantic priors' with 'semantic features' in sec/training.tex. 'Priors' is jargon that may confuse non-experts.
- [writing] Replace 'strict semantic alignment' with 'strong alignment with semantic features' in sec/training.tex. Avoid intensifiers that add no meaning.
- [writing] Replace 'diffusion-friendly' with 'suitable for diffusion modeling' in sec/training.tex. Avoid creating new jargon.
- [writing] Replace 'geometric robustness' with 'geometric stability' in sec/experiment.tex. 'Robustness' is vague.

paper_reviewer_logical_consistency

Verdict: minor_revision
The paper presents a logically consistent argument for the necessity of high-compression VAEs and the proposed solutions (GSC, channel expansion, semantic alignment). The causal chain from problem (high compression leads to detail loss and poor diffusability) to solution (GSC for detail, alignment for diffusability) is well-structured.

However, there are minor logical gaps in the justification of specific design choices and the interpretation of results:

1. **Justification for Removing KL Loss:** The argument that KL loss is removed because "target semantic features are not necessarily Gaussian-distributed" is logically weak. The semantic alignment loss ($\mathcal{L}_{align}$) uses cosine similarity and distance matrix similarity, which do not assume a Gaussian distribution. The conflict between KL loss and semantic alignment is likely due to the *optimization objective* (forcing a specific prior vs. matching a feature manifold) rather than the *distributional assumption* of the target features. The paper should refine this argument to focus on the optimization conflict rather than the distributional mismatch.

2. **DiT Efficiency Claim:** The claim that "channel expansion does not compromise DiT training efficiency" because of a linear projection is partially misleading. While the self-attention complexity in the DiT depends on sequence length (L) and not channel dimension (C), the linear projection layer itself has a computational cost proportional to C. The paper should clarify that the *transformer* complexity is invariant to C, but the *projection* cost scales with C, and that this trade-off is acceptable given the benefits.

3. **Benchmark Comparison Clarity:** In the text describing Table 1 (tab:text_bench), the claim that Qwen-Image-VAE-2.0-f16c128 surpasses "all evaluated f8 VAEs" is technically correct based on the data (0.9617 > 0.9546). However, the table also includes FLUX.2-dev (f16) with NED 0.9535. The text could be clearer in distinguishing that the comparison is specifically against f8 baselines to avoid any confusion with the f16 baseline, even though the logic holds.

These issues are minor and do not invalidate the core contributions but require clarification to ensure the logical consistency of the arguments is fully robust.

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paper_reviewer_overreach

Verdict: minor_revision
The paper exhibits several instances of over-claiming where the conclusions extend beyond the immediate evidence provided in the tables and text.

First, in Section 4.2.2 ("NED for Text Fidelity"), the authors state: "To the best of our knowledge, this is the first f16 autoencoder to achieve text fidelity exceeding f8 VAEs." This claim is an overreach. Table 2 explicitly lists FLUX.2-dev (f16c128) with an NED of 0.9535, which is statistically indistinguishable from the proposed Qwen-Image-VAE-2.0-f16c128 (0.9617) without error bars or significance testing. By ignoring the near-parity with a direct competitor in the same compression class, the paper exaggerates the uniqueness of its achievement. The claim should be tempered to reflect that it matches or slightly exceeds the state-of-the-art, rather than being the definitive "first" to surpass f8 baselines.

Second, the Conclusion (Section 6) asserts that the work "demonstrates a clear technical path to resolving the fundamental tripartite trade-off." This is a strong over-claim. While the model improves the balance between compression, fidelity, and diffusability, it does not "resolve" the trade-off. Table 1 shows that while the proposed f16 model has better reconstruction than some f8 models, it still lags significantly in generation quality (gFID 10.29 vs. 0.55 for FLUX.1-dev f8). The trade-off remains; the model simply shifts the operating point. The language should be adjusted to "mitigating" or "improving the Pareto frontier of" rather than "resolving."

Finally, Section 5.1 claims that removing the KL loss is the primary driver for "enhanced semantic alignment" and a "more flexible latent space." This causal link is over-simplified. The paper introduces a new semantic alignment loss ($\mathcal{L}_{align}$) simultaneously with removing the KL term. Without an ablation study isolating the removal of KL from the addition of $\mathcal{L}_{align}$, it is impossible to attribute the success solely to the removal of the KL constraint. The paper over-attributes the benefit to the architectural simplification rather than the new objective function.

- [writing] The claim that Qwen-Image-VAE-2.0-f16c128 is the 'first f16 autoencoder to achieve text fidelity exceeding f8 VAEs' (Sec 4.2.2) is an overreach. Table 2 shows FLUX.2-dev (f16c128) achieving NED 0.9535, which is extremely close to the proposed model's 0.9617. The paper fails to discuss this near-parity or provide statistical significance testing to justify the 'first' and 'surpassing' superlatives.

- [writing] The conclusion states the model 'resolves the fundamental tripartite trade-off' (Sec 6). This is an over-claim. While the model improves the Pareto frontier, the data in Table 1 shows that f8 baselines (e.g., FLUX.1-dev) still achieve superior generation metrics (gFID 0.55 vs 10.29) and competitive reconstruction. The paper does not demonstrate a true resolution of the trade-off, only a shift in the balance.

- [science] The assertion that 'removing KL loss... achieves a more flexible latent space' (Sec 5.1) is presented as a definitive causal finding without ablation data isolating the KL removal from the semantic alignment loss. The paper conflates the effects of the new loss function with the removal of the prior, over-attributing the success to the architectural change alone.

paper_reviewer_safety_ethics

Verdict: minor_revision
The manuscript presents a technical report on Qwen-Image-VAE-2.0, focusing on high-compression image reconstruction. From a safety and ethics perspective, the primary concerns revolve around data provenance, privacy, and the potential for dual-use in generating deceptive content, particularly given the model's enhanced text-rendering capabilities.

Data Privacy and Consent: In 'sec/data.tex', under the "Human inspection" subsection, the authors state that they "manually prune samples" to ensure quality. However, the paper lacks any mention of Institutional Review Board (IRB) approval or ethical review processes for this human-in-the-loop data curation. Furthermore, the "Data Collection" section mentions scaling to "billions of images" and curating a "specialized document corpus" including "exam papers" and "financial reports." The paper does not specify the licensing or consent status of these documents. If these documents contain personally identifiable information (PII) or sensitive data (e.g., student records, proprietary financial data), their inclusion in a public training set poses significant privacy risks. The authors must explicitly clarify the source of these documents and confirm that they are either public domain, properly licensed, or that PII has been rigorously scrubbed.

Copyright and Synthetic Data: The "Data Synthesis" section in 'sec/data.tex' describes a pipeline that renders text onto backgrounds "randomly sampled from general-domain images." The paper fails to disclose the source of these background images. If these backgrounds are derived from copyrighted datasets (e.g., LAION, Common Crawl) without proper licensing or attribution, the resulting model could inherit copyright liabilities. Additionally, the use of synthetic data to train a model capable of high-fidelity text reconstruction raises concerns about the potential for generating convincing forgeries or disinformation. While the paper focuses on reconstruction, the ability to perfectly reconstruct text from compressed latents could be misused to create deepfakes of documents.

Recommendations: 1. **Ethical Disclosure:** Add a dedicated "Ethical Considerations" section or expand the "Data" section to explicitly state whether IRB approval was obtained for human inspection and how consent was handled for the document corpus. 2. **Data Provenance:** Provide specific details on the licensing and source of the "general-domain images" used for background synthesis in the data pipeline. 3. **Sensitive Content:** Clarify the handling of sensitive document types (exams, financial reports) to ensure no private or confidential information is present in the benchmark or training data. 4. **Dual-Use Discussion:** Briefly discuss the potential dual-use risks of high-fidelity text reconstruction, particularly regarding document forgery, and any mitigation strategies employed (e.g., watermarking, usage restrictions).

Without these clarifications, the paper's claims regarding the safety and ethical robustness of the dataset and model remain unsubstantiated.

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- [writing] Data Provenance: Provide specific details on the licensing and source of the "general-domain images" used for background synthesis in the data pipeline.
- [writing] Dual-Use Discussion: Briefly discuss the potential dual-use risks of high-fidelity text reconstruction, particularly regarding document forgery, and any mitigation strategies employed (e.g., watermarking, usage restrictions). Without these clarifications, the paper's claims regarding the safety and ethical robustness of the dataset and model remain unsubstantiated.

paper_reviewer_scientific_evidence

Verdict: full_revision
The scientific evidence supporting the central claims of Qwen-Image-VAE-2.0 is currently insufficient to validate the "state-of-the-art" assertions, primarily due to a lack of statistical rigor and potential confounding variables in the experimental design.

First, the quantitative results presented in **Table 1** (sec/experiment.tex) and **Table 2** (sec/experiment.tex) report point estimates (PSNR, SSIM, NED, IS, gFID) without any measure of variance (standard deviation) or statistical significance testing. For the OmniDoc-TokenBench, which contains only ~3,000 images, the margin of error for metrics like NED could be substantial. Without confidence intervals or p-values, it is impossible to determine if the reported improvements (e.g., NED 0.9617 vs 0.9546) are statistically significant or merely artifacts of random sampling. The claim of "superior" performance requires effect size analysis to be scientifically robust.

Second, the evaluation of "Diffusability" in **Table 1** is methodologically compromised. The authors compare generation quality (IS/gFID) across models with vastly different latent channel dimensions (64, 128, 192) and, critically, different downstream DiT architectures (SiT-XL/2 for f8 vs. SiT-XL/1 for f16/f32). The change in DiT architecture introduces a massive confounding variable; the observed performance gains could be attributed to the specific capacity or inductive biases of the SiT-XL/1 model rather than the quality of the VAE latent space. A valid comparison requires training the *same* DiT architecture on all VAE latents or rigorously controlling for model capacity.

Third, the construction of the **OmniDoc-TokenBench** (sec/bench.tex) relies entirely on PP-OCRv5 for both the reference "ground truth" and the evaluation metric. The authors argue that OCR errors cancel out, but this assumes the OCR model performs identically on the original and reconstructed images. If the original image contains artifacts that cause the OCR to hallucinate a specific character, and the VAE reconstruction smooths this out (or vice versa), the NED metric will be biased. The lack of a human-annotated subset to validate the OCR-based ground truth undermines the reliability of the text-fidelity claims, especially for the "SOTA" assertion in text-rich scenarios.

Finally, the ablation study for **Global Skip Connections (GSC)** (sec/model.tex, Figure 2) is limited to a single configuration (f16c64). It does not isolate the contribution of GSC from the increased channel dimension (C) in the larger models (f16c128, f32c192). Without ablations showing that GSC provides a consistent boost across different channel widths and compression ratios, the claim that GSC is a key architectural innovation driving the SOTA results remains speculative.

To proceed, the authors must provide statistical significance tests for all benchmark results, re-run the diffusability experiments with a controlled DiT architecture, and validate the OmniDoc-TokenBench metrics against a human-annotated gold standard.

- [science] The claim of 'state-of-the-art' performance in Table 1 lacks statistical validation. No standard deviations, confidence intervals, or significance tests (e.g., t-tests) are reported for the PSNR/SSIM/NED metrics. Given the small benchmark size (~3K for OmniDoc), effect sizes must be quantified to rule out random variance.
- [science] The 'Diffusability' evaluation (Table 1, Generation columns) is methodologically flawed. Comparing IS/gFID across models with different latent dimensions (c64 vs c128 vs c192) and different DiT architectures (SiT-XL/2 vs XL/1) without controlling for model capacity or training steps introduces severe confounding variables. The observed improvements may stem from architectural differences rather than latent space quality.
- [science] The OmniDoc-TokenBench construction relies on PP-OCRv5 for both ground truth and evaluation. If the OCR model fails on the original image (e.g., due to blur or complex layout), the 'ground truth' string is incorrect, artificially inflating the NED score for the reconstruction if it happens to match the OCR error. The paper claims this cancels bias, but systematic OCR failures on specific document types (e.g., dense tables) could skew results. A human-annotated subset validation is required.
- [science] The ablation study for Global Skip Connections (GSC) in Figure 2 is insufficient. It only compares NSC, LSC, and GSC on a single f16c64 model trained from scratch. It does not demonstrate that GSC is the primary driver of the final SOTA performance across the full suite (f16c128, f32c192) or if the gains are merely due to the increased channel dimension (C) alone.

paper_reviewer_statistical_analysis

Verdict: full_revision
The statistical rigor of the evaluation in 'sec/experiment.tex' is insufficient to support the strong claims of "state-of-the-art" performance.

First, **lack of uncertainty quantification** is a critical omission. Tables 1 and 2 present point estimates for PSNR, SSIM, and NED without any measure of variance (e.g., standard deviation) or confidence intervals. In reconstruction tasks, performance can fluctuate based on the specific subset of images sampled. Without reporting the standard deviation over multiple random seeds or bootstrap confidence intervals, it is impossible to determine if the reported margins (e.g., the 0.0071 SSIM gain in Table 2) are statistically significant or within the noise floor of the evaluation. The claim of superiority requires statistical significance testing (e.g., paired t-tests or Wilcoxon signed-rank tests) across the benchmark samples.

Second, the **validity of the NED metric** relies on an unverified assumption. Equation 1 defines NED based on the output of a single OCR model (PP-OCRv5). The authors assert that ”biases largely cancel” when comparing original and reconstructed images. However, OCR models have non-trivial error rates and stochastic behaviors. If the OCR model fails to recognize text in the original image due to noise, or if it hallucinates characters, the NED calculation becomes confounded. The paper lacks a sensitivity analysis or an error propagation estimate to demonstrate that the observed NED improvements are robust to the inherent variance of the OCR engine. A control experiment measuring the OCR model’s self-consistency (NED of an image against itself after OCR) would be necessary to establish a baseline noise floor.

Finally, the **sample size justification** for OmniDoc-TokenBench is missing. The benchmark consists of ~3,000 images. While this is a substantial number, the paper does not provide a power analysis or statistical justification for this size relative to the effect sizes observed. For the f32 compression tier, where baseline performance varies drastically (NED 0.07 to 0.57), the variance is high. Without reporting the distribution of scores (e.g., via box plots or error bars) or conducting a power analysis, the robustness of the conclusions regarding the f32c192 model’s superiority remains statistically unproven.

To proceed, the authors must re-run evaluations to report mean ± standard deviation, perform significance testing against baselines, and provide a sensitivity analysis for the OCR-based metric.

- [science] The paper reports single-point metrics (PSNR, SSIM, NED) without confidence intervals, standard deviations, or statistical significance testing. Given the claim of ’state-of-the-art’ performance, statistical validation (e.g., bootstrapped CIs or paired t-tests) is required to distinguish signal from noise, especially for marginal gains in Table 1 and 2.
- [science] The NED metric calculation (Eq. 1) relies on a single OCR model (PP-OCRv5) without reporting its own error rate or variance. The claim that ’biases largely cancel’ is unverified; a sensitivity analysis or error propagation estimate is needed to ensure NED differences reflect VAE performance rather than OCR stochasticity.
- [science] The benchmark construction (Sec 2.2) involves deterministic filtering and deduplication but lacks a statistical power analysis or justification for the sample size (~3K). It is unclear if this sample size is sufficient to detect the reported effect sizes with adequate power, particularly for the f32 compression tier where baselines vary widely.

paper_reviewer_text_formatting

Verdict: minor_revision
The LaTeX source code for the Qwen-Image-VAE-2.0 Technical Report contains several text formatting and hygiene issues that require attention before final compilation.

First, the preamble in ‘colm2024_conference.tex‘ is cluttered with duplicate package imports. Specifically, ‘booktabs‘, ‘enumitem‘, ‘makecell‘, ‘array‘, ‘longtable‘, ‘inputenc‘, ‘fontenc‘, ‘tcolorbox‘, and ‘wrapfig‘ are declared multiple times. While LaTeX often handles this gracefully, it is best practice to remove these redundancies to prevent potential conflicts and ensure a clean compilation log.

Second, there is a critical file naming mismatch. In ‘sec/bench.tex‘, the code references ‘\includegraphics{pics/OmniDoc-TokenBench-1.pdf}‘ (with a hyphen and capital ’D’). However, the provided file list indicates the actual file is named ‘pics/Omnidoc-TokenBench-1.pdf‘ (mixed case, no hyphen in ’OmniDoc’). This discrepancy will cause a compilation error. The filename in the source must be updated to match the filesystem exactly.

Third, the usage of cross-referencing is inconsistent. The document loads ‘cleveref‘ and defines custom formats, yet the text frequently uses manual ”Figure~\ref{...}” or ”Table~\ref{...}” instead of ‘\cref{...}‘ or ‘\Cref{...}‘. For instance, ‘sec/experiment.tex‘ uses ”Table~\ref{tab:main_bench}” while ‘sec/model.tex‘ uses ”Table~\ref{tab:vae_configs}”. Consistent use of ‘cleveref‘ would automatically handle capitalization and pluralization, improving the professional appearance of the document.

Finally, there are minor LaTeX syntax issues in the mathematical expressions. In ‘sec/-training.tex‘, the function ‘ReLU‘ is used in equations without being defined as a math operator (e.g., ‘\operatorname{ReLU}‘ or ‘\mathrm{ReLU}‘), which renders it in italic math font rather than upright text. Similarly, in ‘sec/bench.tex‘, the definition of the NED metric uses ‘d_{\mathrm{edit}}‘ but the surrounding text formatting could be tightened for better readability.

Addressing these formatting, naming, and syntax issues will ensure the paper compiles without errors and adheres to high-quality typesetting standards.

- [writing] Remove duplicate package imports in the preamble. ’booktabs’, ’enumitem’, ’makecell’, ’array’, ’longtable’, ’inputenc’, ’fontenc’, ’tcolorbox’, and ’wrapfig’ are loaded multiple times, which can cause compilation warnings or conflicts.
- [writing] Fix inconsistent figure file naming. The text in ’sec/bench.tex’ references ’pics/OmniDoc-TokenBench-1.pdf’ (hyphenated), but the provided file list shows ’pics/Omnidoc-TokenBench-1.pdf’ (mixed case). Ensure the filename in the LaTeX source matches the actual file on disk exactly.
- [writing] Standardize citation formatting. The manuscript inconsistently uses ’Figure~ ef{...}’ and ’Fig.~ ef{...}’. Additionally, ’sec/experiment.tex’ uses ’Table~ ef{...}’ while ’sec/-model.tex’ uses ’Table~ ef{...}’ but the caption in ’sec/experiment.tex’ uses ’Table’ while the text refers to ’Table’ inconsistently with the ’cleveref’ setup. Ensure ’cleveref’ is used consistently or standard ’ref’ is used throughout.

- [writing] Correct LaTeX math mode spacing and syntax. In 'sec/training.tex', the equation for L_mcos uses 'ReLU' without a backslash (should be \mathrm{ReLU} or \operatorname{ReLU}). In 'sec/bench.tex', the equation for NED uses 'd_{\mathrm{edit}}' but the text refers to 'Levenshtein distance' without consistent formatting. Ensure all math operators are properly defined.

paper_reviewer_writing_quality

Verdict: minor_revision
The manuscript presents a technically dense and well-structured report on Qwen-Image-VAE-2.0. The overall flow is logical, moving from the problem statement to architectural innovations, data strategies, and comprehensive evaluations. The writing is generally clear and professional, effectively communicating complex concepts regarding high-compression VAEs and semantic alignment.

However, there are several specific instances of grammatical errors and awkward phrasing that detract from the polish of the paper. In ‘sec/training.tex‘, the authors write "gradually loose the alignment margins." Here, "loose" is used as a verb, but the correct verb form is "loosen." This is a clear grammatical error that must be fixed.

In ‘sec/experiment.tex‘, under the "Performance of Text Rendering" section, a comma splice occurs: "We compute NED on raw OCR outputs without text normalization, while minor spacing artifacts from OCR may inflate edit distance, the evaluation pipeline is applied consistently..." The clause following the comma is independent and requires a stronger conjunction or punctuation (e.g., a semicolon or "however") to connect it properly to the preceding thought.

Additionally, in ‘sec/introduction.tex‘, the phrasing "despite its large channel dimension" creates a slight ambiguity regarding the antecedent, as the previous sentence discusses "large-channel VAEs" in the plural. Adjusting this to "despite the large channel dimensions" would improve precision. Finally, in ‘sec/model.tex‘, the caption for Figure 1 uses the phrasing "abbreviation of," which is less idiomatic than "abbreviation for" or "stands for."

Addressing these specific writing issues will significantly enhance the readability and professional quality of the manuscript without altering the scientific content.

- [writing] In sec/training.tex, the phrase 'gradually loose the alignment margins' contains a grammatical error. 'Loose' is an adjective; the verb form required here is 'loosen'. This should be corrected to 'gradually loosen the alignment margins'.

- [writing] In sec/experiment.tex, the sentence 'while minor spacing artifacts from OCR may inflate edit distance, the evaluation pipeline is applied consistently...' is a comma splice. It joins two independent clauses with only a comma. Please insert a semicolon, a period, or a conjunction (e.g., '...edit distance; however, the evaluation...').

- [writing] In sec/introduction.tex, the phrase 'despite its large channel dimension' appears in the final paragraph of the introduction. The antecedent for 'its' is slightly ambiguous given the previous sentence discusses 'Qwen-Image-VAE-2.0' (singular) and

’large-channel VAEs’ (plural). Rephrase to ’despite the large channel dimensions’ for clarity.

- [writing] In sec/model.tex, the caption for Figure 1 contains the phrase ’S2C is the abbreviation of Space to Channel module.’ This is slightly awkward. It should be ’S2C is the abbreviation for the Space-to-Channel module’ or ’S2C stands for Space-to-Channel’.